Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/19/2025
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	02/15/2021	**End Date:**	09/30/2033
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2033 per C. Ribeiro/JSC (Ed., 3/21/23). NOTE: End date changed to 02/14/2023 per C. Ribeiro/JSC (Ed., 4/7/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Karanjia, Rustum M.D., Ph.D. (Doheny Eye Institute) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Martin, Bryn Ph.D. (Alcyone Therapeutics Inc.) Sadda, Srinivas M.D. (Doheny Eye Institute) Young, Millennia Ph.D. (NASA Johnson Space Center)		

Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The proposed study will identify long-term health consequences of spaceflight on the structure and function of the eye and brain. Experiments will be performed with crewmembers before, during, and after six-month spaceflight missions; post-flight surveillance will span five years. In another arm we will characterize the structural and functional changes to the eye and brain, including cognitive changes, in crewmembers who have participated in past missions, calling them back several years after their most recent flights. The data delivered from this study will provide NASA with information necessary for completing the risk assessment for future interplanetary expeditions. This proposal primarily addresses the following two Human Research Roadmap Gaps: 1) We do not know the etiological mechanisms and contributing risk factors for ocular structural and functional changes seen in-flight and post-flight and 2) Determine if exposure to long-duration spaceflight leads to neural structural alterations and if this remodeling impacts cognitive and functional performance. Ed. Note (2/29/24): This project has been combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The protocols and aims from all four PIs have been integrated into a revised proposal that was delivered to NASA Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Element Scientists in January 2022. For information on the four related investigations, see: • The Long-Term Consequences of Spaceflight on Brain and Eye Health (PI: Bershad). • Long-Term Brain Structural and Functional Consequences of Spaceflight (PI: Basner). • Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight (PI: Macias). • Recovery Timeline of Spaceflight-Induced Central Nervous System Changes (PI: Seidler).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterizing the long-term effects of Spaceflight Associated Neuro-ocular Syndrome (SANS) will provide NASA with information necessary for completing the risk assessment for future interplanetary expeditions. Furthermore, the unique nature of this project may lead to the adoption of new hardware, software, or protocol elements in clinical practice, benefiting patients on Earth. For example, novel methods to be used in this study are augmenting capabilities of existing technologies, e.g., deformation mapping for ocular Magnetic Resonance Imaging (MRI).
Task Progress:	Major accomplishments during this reporting period for this study include: • Began data collection and collected data from two crewmember participants • Began recruitment of control participants The PI team is also holding bi-weekly meetings with the “Long-Term Health” (LTH) Co-Principal Investigator (PI) teams to coordinate ongoing data collection and analysis efforts for the LTH Call Back study. Ongoing forward work with Research Operations and Integration (ROI) for “LTH Call Back” includes: • The LTH Call Back study will continue recruitment and data collection, including During recruitment 2 astronauts and 4 control TSS participants during the next reporting period. BACKGROUND Spaceflight-associated neuro-ocular syndrome (SANS) describes structural and functional alterations of the eye that develop in many astronauts during long-duration spaceflight missions (>6 month). Understanding the consequences of the structural and functional changes to the eye and brain that endure long after return from spaceflight is essential for future exploration-class spaceflight missions. PURPOSE The primary purpose of this project is to characterize the structural and functional changes to the eye and brain, including cognitive changes, in crewmembers who have participated in past missions, calling them back several years after their most recent flights. METHODS Outcomes from astronauts who had previously participated in ≥6-month missions (n=20) will be compared to control subjects who do not have standard-duration spaceflight experience (n=20). In addition to characterizing the effects observed for individual variables, all data collected during the study will be analyzed using an integrative approach to identify relationships across ocular measures, brain measures, individual crewmembers, and time. Ocular structure and function will be evaluated using magnetic resonance imaging (MRI) and 3D structural analysis, optical coherence tomography (OCT), angiography, electroretinography (ERG), visual field analysis, pneumotonometry (intraocular pressure), visual field analysis, and optical biometry. In collaboration with the integrated project team, MRI will be used to assess brain structural changes, and functional MRI (fMRI), and a series of cognitive batteries will be used to evaluate brain function. To better understand the neurophysiological mechanisms underlying anticipated changes, biomarkers associated with specific structural and functional brain alterations will be measured in blood samples. To complement the research measures, medically relevant test results already collected by the Space Medicine Operations Division will be shared.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Carter KJ, Basner M, Bershad EM, Seidler RD, Stahn AC, Laurie SS, Macias BR. "Investigating long-term structural and functional changes to the eye and brain after spaceflight." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025